INCH-POUND
MIL-DTL-24643/9G
27 November 2023
SUPERSEDING
MIL-DTL-24643/9F
1 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 600 VOLTS, TYPE LSMRI

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction (Non-Watertight)

First - Copper conductor, coated or uncoated, class K stranding (see table I for size).

Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.

Third - Thermoset insulation. Minimum average wall thickness of 0.020 inch for 16 AWG and 0.015 inch for 20 AWG. Standard identification code applied by method 3.

Fourth - The specified number of conductors shall be cabled together in accordance with 3.4.4 of MIL-DTL-24643.

NOTE 1: The black wire shall be printed in accordance with 3.4.10 of MIL-DTL-24643, without the jacket type information. The year of manufacture may be printed on the black or white wire of the cable. The printed characters may be parallel or perpendicular to the longitudinal axis of the wire and shall be sized so as to be legible without magnification.

NOTE 2: Manufacturer's identification tape (see 3.4.9 of MIL-DTL-24643) is not applicable.

Distribution Statement A. Approved for public release: distribution is unlimited.

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TABLE I. Details.

Military part no. M24643/9	Type and size	Conductors		Insulated conductor diameter (inch)		Conductor resistance (ohms)
		No.	AWG	min.	max.	
-01UN	LSMRI-D-1	2	20	0.068	0.075	11.05
-02UN	LSMRI-D-2½	2	16	0.092	0.102	4.36
-03UN	LSMRI-T-2½	3	16	0.092	0.102	4.36

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	See table I
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor, 20 AWG	1,000
Conductor to conductor, 16 AWG	1,500
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to conductor (all sizes)	100
Conductor continuity	No failure

Group A:

Visual and dimensional	No failure

Group B:

Cold bending (cable)	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150

Group C:

Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (percent of unaged, min.)	75

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Elongation (percent of unaged, min.)	75
Permanence of printing (insulation) (cycles, min.)	25
Heat distortion (percent of unaged, max.)	30

Group D:

Flame propagation (cable)	Not applicable

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Acid gas equivalent (percent, max.)	
Insulation	18
Halogen content (percent, max.)	
Insulation	0.2
Smoke index (max.)	
Insulation	45
Toxicity index (max.)	
Insulation	1.5
Electrical moisture absorption	No failure
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSMRI-D-1	600
LSMRI-D-2½	600
LSMRI-T-2½	500

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-025)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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